

Scalability & Future Plan

Date	15 July 2026
Team ID	XXXX
Project Name	A Comprehensive Measure of Well-being
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Application is designed for single user prediction at a time.	Cannot handle multiple users simultaneously, limiting accessibility.	Medium
2	Model accuracy depends on the quality and size of the dataset.	Prediction accuracy may reduce with insufficient or outdated data.	High
3	Currently limited to prediction using Linear Regression model.	May not capture complex relationships in data compared to advanced models.	Medium
4	No real-time data updates from external sources.	Predictions may not reflect recent changes in development indicators.	Medium
5	Application is not deployed on cloud platform.	Access is limited to local system only.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports single user at a time.	Deploy on cloud platform (Render / AWS / Azure) to handle multiple concurrent users.
2	Data Storage	Uses CSV file for storing dataset.	Migrate to MySQL or PostgreSQL database for efficient management of large datasets and user data.
3	Performance	Predictions generated using Linear Regression model.	Implement advanced ML models (Random Forest, XGBoost) for better accuracy and performance.
4	Security	Basic input validation implemented.	Add user authentication, secure data handling, and input sanitization for better security.
5	Features	Provides prediction and basic visualization.	Add feature importance, trend analysis, country comparison, and downloadable reports.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the web application on cloud platform and improve UI/UX.	1 – 2 Months	Better accessibility, improved user experience, and wider reach.
Phase 2	Integrate database for storing user data and prediction history.	2 – 3 Months	Efficient data management and ability to handle more users.
Phase 3	Implement advanced machine learning models for higher prediction accuracy.	3 – 4 Months	More accurate predictions and better understanding of well-being factors.
Phase 4	Add new features such as country comparison, feature importance, and report generation.	4 – 6 Months	Enhanced insights, detailed analysis, and downloadable reports for users.
Phase 5	Enable real-time data updates using external APIs and automate data preprocessing.	6 – 9 Months	Up-to-date predictions and reduced manual effort in data handling.
Phase 6	Develop mobile application for prediction and analysis.	9 – 12 Months	Mobile accessibility and user convenience on the go.